Thermodynamic State Vector Inventory Discrepancy Evaluator in Complex Ecosystem Simulations

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Sprint 078 – Repository: `https://github.com/pascalranoroarijaona/WebOfLife`

Abstract

We report the formal specification and implementation of the Thermodynamic State Vector Inventory Discrepancy Evaluator within `src/thermodynamics/state_validator.ts` (Sprint 078). This module establishes rigorous First and Second Law thermodynamic compliance across simulated biogeochemical and energy stocks within the Web of Life engine. By encapsulating inventory aggregation and discrepancy analysis into a reusable `evaluateDiscrepancy` method, the simulation monitors mass-energy conservation errors and bounds entropic dissipation during monad state transitions.

1 Introduction and Physical Principles

Ecosystem dynamics within the Web of Life framework are governed by fundamental thermodynamic constraints. Sprint 078 introduces real-time state validation to detect and correct discrepancies between observed ecological states and theoretical expectations.

1.1 First Law of Thermodynamics: Conservation Relations

The conservation of mass and energy across system stocks is formulated as:

$$\sum S_{i,t} = \sum S_{i,t-1} + \sum \Phi_{in,t} - \sum \Phi_{out,t} + \Delta E_{gen} \tag{1}$$

where $S_{i,t}$ denotes stock i at time t, Φ represents boundary fluxes, and ΔE_{gen} accounts for solar or internal generation. Discrepancy vectors δ_i and total discrepancy $\mathcal{D}_{total}$ are evaluated via:

$$\delta_i = S_{curr,i} - S_{exp,i}, \quad \mathcal{D}_{total} = \sum_i |\delta_i| \tag{2}$$

1.2 Second Law of Thermodynamics: Entropy Variation

Irreversible transformations generate entropy proportional to unallocated energy divergence between current and expected states:

$$\Delta S_{entropy} = \kappa \cdot |E_{curr} - E_{exp}| \tag{3}$$

where $\kappa = 0.001\,\mathrm{J} \cdot \mathrm{K}^{-1} \cdot \mathrm{J}^{-1}$ acts as the normalized dissipation coefficient.

2 Software Architecture and Implementation

The validator is implemented in `src/thermodynamics/state_validator.ts` adhering to strict interface contracts:

```
export class StateValidator implements IStateValidator {
  constructor(private tolerance: number = 1e-6) {}

  public evaluateDiscrepancy(currentState: StateVector, expectedState: StateVector): Discrep
    const vectorDiscrepancies = this.aggregateDifferences(currentState, expectedState);
    const totalDiscrepancy = Object.values(vectorDiscrepancies).reduce((acc, val) => acc + M
    const isBalanced = totalDiscrepancy <= this.tolerance;
    const entropyDelta = this.computeEntropyDelta(currentState, expectedState);

    return { timestamp: Date.now(), totalDiscrepancy, vectorDiscrepancies, isBalanced, entro
  }
}
```

3 Test Verification and Results

Testing protocols in `tests/sprint_078.test.ts` verify:

1. **Zero Discrepancy:** Identical vectors yield $\mathcal{D}_{total} = 0$ and $\Delta S = 0$.
2. **Inventory Mismatch:** Perturbations in carbon, nitrogen, phosphorus, and water pools exceed tolerance 10^{-6} and trigger damping feedbacks.
3. **Conservation Enforcement:** Unprovenanced energy inputs trigger First Law boundary violation flags.

4 Conclusion

Sprint 078 successfully equips the Web of Life simulation engine with an automated thermodynamic auditing mechanism, securing mathematical consistency across all ecological monad transitions. Repository access: `https://github.com/pascalranoroarijaona/WebOfLife`.